



Snowflake Secure Data Sharing

DataHive utilizes Snowflake for the warehousing of its data assets.

Snowflake's [Secure Data Sharing](#) feature can facilitate a one-way direct connection between users' Snowflake instances and DataHive's Snowflake instance, with the connection being brokered and managed by Snowflake.

Pre-requisites to receive data from DataHive

- Snowflake instance with a Business Critical tier in the Azure Southeast Asia (Singapore) region.
- A fulfilled Data Request (DR) requesting for data from Trusted Center for Sensors Data (TCS).
- As a DR owner, email to support@datahive.gov.sg with your [7 character Snowflake instance ID](#) (e.g. AB12345) to request for the Snowflake-to-Snowflake secure data share.

How does Secure Data Sharing work?

When tables are written to Snowflake, Snowflake manages the data securely using underlying storage accounts in the respective cloud providers (Azure, AWS, Google Cloud).

Since the data is managed by Snowflake, no actual data is copied or transferred between accounts in a Secure Data Share. Snowflake simply brokers a logical access point to the underlying table/view being shared.

There are a few benefits to this approach of data sharing:

- No data staleness. Users will read from the same underlying source when reading from the Secure Data Share. Changes to the table would be reflected near-instantaneously on the users' end.
- No storage costs. Since there is no data replication, there is no storage costs for users. The only charges are for the compute resources used to query the shared data.

- Less overhead. Secure Data Sharing is brokered and occurs on the Snowflake control plane. There is no need to setup data extraction/ingestion scripts, whitelist firewalls and/or set up file transfer protocols etc.

! INFO

Data Shared is read-only. The shared data cannot be modified by the user.

Direct Share vs Private Listing

Snowflake offers two mechanisms for Secure Data Sharing. They differ mainly in whether they can share data across cloud regions and platforms, and whether data is replicated.

Direct Share is the mechanism DataHive uses today. Snowflake brokers a logical access point to the underlying table/view and **no data is copied or replicated**. Because of this, a direct share can only be consumed by an account in the **same Snowflake region and on the same cloud platform** as the provider. This is why receiving data from DataHive requires a Snowflake account in the Azure Southeast Asia (Singapore) region (see the pre-requisites above).

Private Listing can share data with consumer accounts in **different regions and on different cloud platforms**. To achieve this, Snowflake uses [Cross-Cloud Auto-fulfillment](#) to **automatically replicate** the shared data to the consumer's region/cloud. Snowflake manages the replication, but because a copy of the data is provisioned in each target region, it incurs additional replication and storage costs (borne by the provider).

Aspect	Direct Share	Private Listing
Data replication	None — logical access to the source only	Snowflake replicates the data to the consumer's region/cloud via Cross-Cloud Auto-fulfillment
Cross-region / cross-cloud	Not supported — provider and consumer must be in the same region and cloud platform	Supported

Aspect	Direct Share	Private Listing
Storage footprint & cost	No additional copy; no storage cost for the shared data	A replicated copy is provisioned in each target region, incurring additional storage and replication costs
Data freshness	Reads the live source; changes reflected near-instantaneously	Replicated data is refreshed periodically, so there may be a slight lag behind the source

DataHive's Snowflake instance is hosted in the **Azure Southeast Asia (Singapore)** region, and DataHive shares data only via a **direct share** without replicating data across regions or clouds. Data **providers** are responsible for any cross-cloud replication. The supported sharing scenarios are:

#	From	To	Supported	Reason
1	DataHive	Azure Southeast Asia (Singapore) instance		Same region and cloud — direct share, no replication needed
2	Azure Southeast Asia (Singapore) instance	DataHive		Same region and cloud — direct share, no replication needed
3	AWS Asia Pacific (Singapore) instance	DataHive		Different cloud platform (even in the same geography) — the AWS-side provider replicates its data to DataHive's Azure Southeast Asia (Singapore) Snowflake instance via Cross-Cloud Auto-fulfillment

#	From	To	Supported	Reason
4	DataHive	AWS Asia Pacific (Singapore) instance	✗	Different cloud platform — would require DataHive to replicate data across clouds

Interested in sharing your dataset with DataHive via Secure Data Sharing?

If you have datasets that you believe would benefit whole-of-government users and wish to onboard them to DataHive via Snowflake, please contact us at admin@datahive.gov.sg.